1 Forces, mass, and weight

Key ideas

- A **force** is a push or pull caused by an interaction. It is a vector: give its size and direction. Unit: **newton (N)**.
- **Contact forces:** normal force, friction, and spring force. **Gravity** acts between masses without contact.
- **Mass** measures inertia, is measured in **kg**, and does not change when an object moves to another planet.
- **Weight** is gravitational force, is measured in **N**, and changes when the local gravitational field changes.
- Near Earth, weight points downward. A spring balance measures force; for a hanging object at rest, its reading equals the object's weight.

Equations to remember

$F_g = mg$ $\quad m = F_g / g$ $\quad g = F_g / m$

Earth: $g = 9.81\ \text{N/kg} = 9.81\ \text{m/s}^2$

Use kilograms before substituting. **1 kg = 1000 g.** Force and mass need different units.

Check yourself

Q1. A 60.0 kg person stands on Earth. Calculate their weight, including direction.

Q2. A 4.00 kg object weighs 14.8 N on another planet. Find the gravitational field strength there. What happens to its mass?

Q3. Explain the difference between mass and weight. If you moved to the Moon, which quantity would change?

Review routine

Cover the notes, explain the key ideas aloud, and attempt each question. Show the equation, substitution, unit, and direction where needed. Check answers on pages 8–9 only after trying.

2 Universal gravitation

Key ideas

- Any two masses attract. The forces on the two objects have **equal magnitude and opposite directions**, even if the masses differ.
- More mass means a larger force. Greater separation means a smaller force, following an **inverse-square** relationship.
- Measure **r between the centres** of the objects. For a spherical planet at altitude h, use **r = R + h**, where R is its radius.
- **G** is the universal gravitational constant. **g** is the local gravitational field strength; they are different quantities.

Equations to remember

$$F_g = \frac{Gm_1m_2}{r^2} \qquad g = \frac{GM}{r^2}$$

$$G = 6.67 \times 10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2$$

Use kg and metres. The planetary field equation applies outside a spherical planet of mass M.

Changes you should recognise

- Double one mass → **2F**. Double both masses → **4F**.
- Double separation → **F/4**. Triple separation → **F/9**.
- Halve separation → **4F**. Always square the distance factor.

Check yourself

Q4. Two objects attract with force F. Their masses stay unchanged and their centre-to-centre separation triples. Find the new force.

Q5. At a height equal to a planet's radius above its surface, what fraction of the surface gravitational field remains? Show the distance from the centre.

Q6. Two golf balls each have mass 0.046 kg. Their centres are 0.10 m apart. Calculate their gravitational attraction.

---

3 Falling objects and gravity graphs

Key ideas

- With negligible air resistance, objects released from the same height with the same initial vertical velocity fall together, regardless of mass.
- A heavier object feels more gravitational force, but its greater inertia means the same free-fall acceleration.
- Paper falls slowly in air because air resistance has a large effect relative to its weight. A book beneath it can shield it from much of that resistance.
- A horizontally launched ball and a dropped ball land together if their release heights and initial vertical velocities match and air resistance is negligible.

Graphs to remember

> Vertical axis F_g; horizontal axis m:
> slope = $\Delta F_g / \Delta m = g$ units: N/kg
>
> Vertical axis F_g; horizontal axis $1/r^2$:
> slope = Gm_1m_2 fixed masses

Force against distance r is curved. Force against **$1/r^2$** is straight in the ideal model. Use well-separated points on a best-fit line to calculate slope.

Check yourself

Q7. A book and a flat sheet of paper are dropped separately, then with the paper resting entirely on the book. Explain why the outcomes differ.

Q8. One ball is dropped while another is launched horizontally at the same instant and height. Both start with zero vertical velocity. Which lands first? State the assumption.

Q9. A graph has weight F_g vertically and mass m horizontally. What does its slope represent, and what are its units? What would the slope represent if the axes were reversed?

---

4 Static friction, sliding, and normal force

Key ideas

- **Static friction** acts when the surfaces are not sliding. It adjusts to oppose the tendency to slip, up to a maximum.
- **Kinetic friction** acts while surfaces slide and opposes their relative sliding at the contact.
- Starting motion often requires more force than maintaining steady sliding: maximum static friction commonly exceeds kinetic friction.
- The **normal force, F_N**, acts perpendicular to the contacting surface. On a level floor it equals mg only when vertical forces balance and there are no other vertical forces.
- At rest or at constant velocity, the net force is zero. A moving object does not automatically need an unbalanced forward force.

Equations to remember

> Sliding: $F_{fr} = \mu_k F_N$
> Static: $0 \le F_{static} \le \mu_s F_N$

Both coefficients are **dimensionless**. Static friction equals its maximum only at the threshold of slipping.

Force diagram reminder

For a box pulled horizontally across a level floor: normal force upward, weight downward, pull forward, friction backward. Draw only forces acting **on the box**.

Check yourself

Q10. A box stays at rest under a 5 N horizontal push. No other horizontal applied force acts. What are the size and direction of its static friction? Must it equal the maximum static friction?

Q11. A box slides straight across a level floor at constant speed under a horizontal push. Describe the vertical and horizontal force balances.

Q12. A string pulls a box at an upward angle across a level floor, with no vertical acceleration. Is F_N equal to mg? Explain how the pull affects the normal force.

---

5 Friction calculations and fair tests

Key ideas

- First identify sliding or static contact. Draw the forces and find F_N before choosing a friction equation.
- For horizontal steady sliding, a horizontal pull equals kinetic friction. Under these conditions, F_N equals the total weight.
- For the same dry surfaces, kinetic friction is approximately proportional to normal force. Changing apparent contact area alone has little effect in this model.
- To test load: add blocks to the same bottom block. To test contact area: rearrange the same total load. Change one variable at a time.

Equations and graph

$F_{fr} = \mu_k F_N$ $\mu_k = F_{fr} / F_N$

Graph F_{fr} vertically against F_N horizontally:

slope = μ_k no units

Keep the pull horizontal and the speed steady; zero the force meter and repeat readings. Use the steady sliding force, not the brief peak at the start.

Check yourself

Q13. A wooden box weighs 450 N and slides steadily on a level floor. A horizontal push keeps it moving. If $\mu_k = 0.30$, calculate the required push.

Q14. A box weighing 40 N requires a 2.5 N horizontal pull to slide steadily. An 80 N load is added on top. Find μ_k and the new steady pull, assuming the same contact surfaces.

Q15. In a dry sliding-friction investigation, (a) double the total weight without changing the bottom contact surface, then (b) double apparent contact area while holding total weight fixed. Predict each effect on friction and explain which quantity the graph slope measures.

---

6 Hooke's law and spring graphs

Key ideas

- **Extension is the change in length:** x = loaded length − relaxed length. Do not substitute the whole spring length.
- Within the linear elastic range, doubling the force doubles the extension. The **spring constant k** measures stiffness; a larger k means a stiffer spring.
- The spring's restoring force acts toward its relaxed position: opposite the displacement. A compressed spring pushes back; a stretched spring pulls back.
- Hooke's law may fail if the spring is stretched too far. Rubber bands are not necessarily linear.

Equations and units

> Force magnitudes: $F = kx$ $k = F/x$ $x = F/k$
> Signed restoring force: $F_s = -kx$
> 1 cm = 0.01 m 1 N/cm = 100 N/m

A graph of applied force vertically against extension horizontally has **slope k**. With metres on the horizontal axis, the slope has units N/m. Reversing the axes gives slope 1/k.

Check yourself

Q16. A spring's relaxed length is 10 cm. It is first stretched to 14 cm and then compressed to 6 cm. Taking extension as positive, state each displacement and the restoring-force direction.

Q17. A spring graph shows that 1.0 N produces an extension of 0.75 cm. Find k in N/m. For the same spring, find the extension produced by 4.0 N.

Q18. A 2.5 kg hanging mass extends a spring by 10 cm. What extension will a 5.0 kg mass produce at rest, if the spring remains linear?

---

7 Elastic balances and choosing a model

Key ideas

- A spring or bending stick can be calibrated as a balance: zero it, add known masses, calculate their weights, and record the deformation.
- Plot **force against deformation**. In a linear range, its slope is effective stiffness k_{eff}. A straight line through the origin supports direct proportionality.
- Keep the clamp and overhang fixed. Account for the holder's weight, check that the support returns to zero, and use the calibrated range.
- For a hanging object at rest, upward elastic support balances downward weight: **kx = mg**.

Choose the equation from the situation

Weight in a known field:	$F_g = mg$
Attraction between masses:	$F_g = Gm_1m_2/r^2$
Dry sliding friction:	$F_{fr} = \mu_k F_N$
Linear spring deformation:	$F = kx$
Hanging spring at rest:	$kx = mg$
Unbalanced forces:	$F_{net} = ma$

For acceleration, choose a positive direction and add forces with signs. Opposing horizontal forces subtract.

Check yourself

Q19. An elastic balance has k_{eff} = 200 N/m. An unknown load causes a depression of 3.00 cm at rest. Find the load's weight and mass on Earth.

Q20. Describe how you would calibrate a clamped metre-stick balance to find an unknown object's mass. Give one control and one check of reliability.

Q21. A 5.0 kg desk is pushed horizontally with 157 N while friction is 123 N in the opposite direction. Find its net force and acceleration.

Final check

Can you explain each symbol, choose the model, convert to SI units, and state a sensible answer with units? Review any question you could not solve without looking at the notes.

8 Answers: questions 1–12

Mass and weight · Q1–3

1. $F_g = mg = (60.0)(9.81) =$ **589 N downward** (588.6 N before rounding).

2. $g = F_g/m = 14.8/4.00 =$ **3.70 N/kg**. Its mass remains **4.00 kg**.

3. Mass measures inertia in kg; weight is gravitational force in N. On the Moon, **weight decreases and mass stays the same**.

Universal gravitation · Q4–6

4. $F_{new}/F = (r/3r)^2 = 1/9$. The new force is **F/9**.

5. $r = R + h = 2R$. Therefore $g_{new}/g_{surface} = (R/2R)^2 =$ **1/4**.

6. $F_g = (6.67 \times 10^{-11})(0.046)^2/(0.10)^2 =$ **1.4×10^{-11} N**. Each force points toward the other ball.

Falling and graphs · Q7–9

7. Separately, air resistance slows the paper much more relative to its weight. On the book, the paper is shielded from much of that resistance, so they fall together. With negligible air resistance, both have the same gravitational acceleration.

8. They land together, assuming negligible air resistance. Their initial vertical velocities, release heights, and vertical accelerations are equal; horizontal velocity does not change vertical free fall.

9. The slope is **g in N/kg**. Reversing the axes gives **1/g in kg/N**.

Friction and forces · Q10–12

10. Static friction is **5 N opposite the push**. It is not necessarily at its maximum; it adjusts to keep the box at rest.

11. Vertically, **$F_N = mg$**. Horizontally, **$F_{push} = F_{fr}$**. The net force is zero because velocity is constant.

12. F_N is less than mg. The upward component of the pull supplies part of the support: $F_N + F_{pull,up} = mg$. Find this reduced normal force before calculating friction.

9 Answers: questions 13–21

Friction calculations · Q13–15

13. F_N = 450 N. F_{fr} = (0.30)(450) = 135 N. The push equals friction: **about 1.4 × 10² N forward** (135 N before rounding).

14. μ_k = 2.5/40 = **0.0625** (about 0.063). New F_N = 40 + 80 = 120 N. New pull = (0.0625)(120) = **7.5 N**. The coefficient stays unchanged.

15. (a) Friction approximately **doubles** because F_N doubles. (b) Friction stays approximately **unchanged** because total F_N and the surface pair are unchanged. The slope of F_{fr} against F_N is $\boldsymbol{\mu_k}$, with no units.

Springs · Q16–18

16. Stretch: x = 14 − 10 = **+4 cm**; the restoring force opposes stretching. Compression: x = 6 − 10 = **−4 cm**; the restoring force opposes compression. Both forces point toward the relaxed position.

17. 0.75 cm = 0.0075 m. k = 1.0/0.0075 ≈ **1.3 × 10² N/m**. Using the unrounded k, x = 4.0/(1.0/0.0075) = 0.030 m = **3.0 cm**.

18. For the same spring at rest, $x \propto m$. x_2 = (10 cm)(5.0/2.5) = **20 cm**.

Balances and net force · Q19–21

19. y = 0.0300 m. $F_g = k_{eff}\, y$ = (200)(0.0300) = **6.00 N downward**. m = 6.00/9.81 = **0.612 kg**.

20. Record the unloaded position; add known masses and calculate F_g = mg; measure depression; fit force against depression. Use the graph to find the unknown weight, then divide by g. **Control:** fixed clamp and overhang. **Reliability check:** repeat readings, verify return to zero, or compare with a commercial balance.

21. F_{net} = 157 − 123 = **34 N in the push direction**. a = 34/5.0 = **6.8 m/s² in the push direction**.

Where these questions come from

Condensed and adapted from forces-notes.html: its explanations, worked examples, and selected chapter questions. This review covers the core ideas rather than the full 79-question collection. Source: BC Science Physics 11, Chapter 3, printed pp. 76–100.